

ToS	Type of Service
TSC	Time Sensitive Communication
TSCTSF	Time Sensitive Communication and Time Synchronization function
TSSF	Traffic Steering Support Function
UDP	User Datagram Protocol
UL CL	Uplink Classifier
UMIC	User Plane Node Management Information Container
UP	User Plane
UPF	User Plane Function
URR	Usage Reporting Rule
VID	VLAN Identifier

4 Protocol Stack

4.1 Introduction

The protocol stack for the control plane over the Sxa, Sxb, Sxc, combined Sxa/Sxb, N4 and N4mb reference points shall be as depicted in Figure 4.1-1. Clauses 4.2 and 4.3 further specify the related UDP and IP requirements.

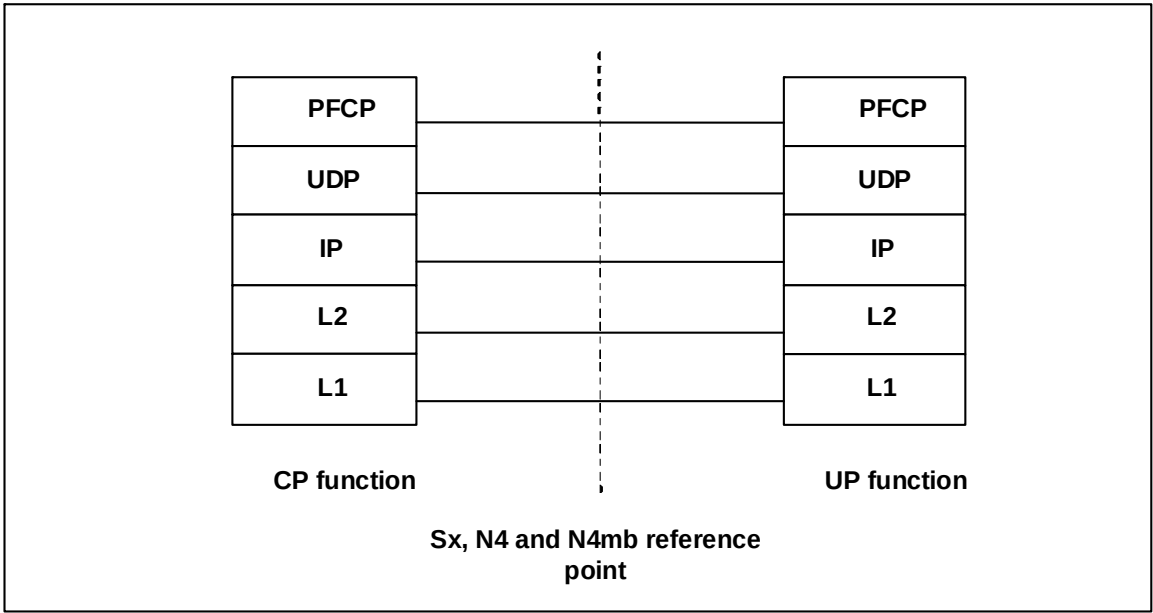


Figure 4.1-1: Control Plane stack over Sxa, Sxb, Sxc, combined Sxa/Sxb, N4 and N4mb

The protocol stack for the user plane over the Sxa, Sxb and N4reference points (see clause 5.3) shall be as depicted in Figure 4.1-2. 3GPP TS 29.281 [3] further specifies the related GTP-U, UDP and IP requirements. Both IPv4 and IPv6 shall be supported.

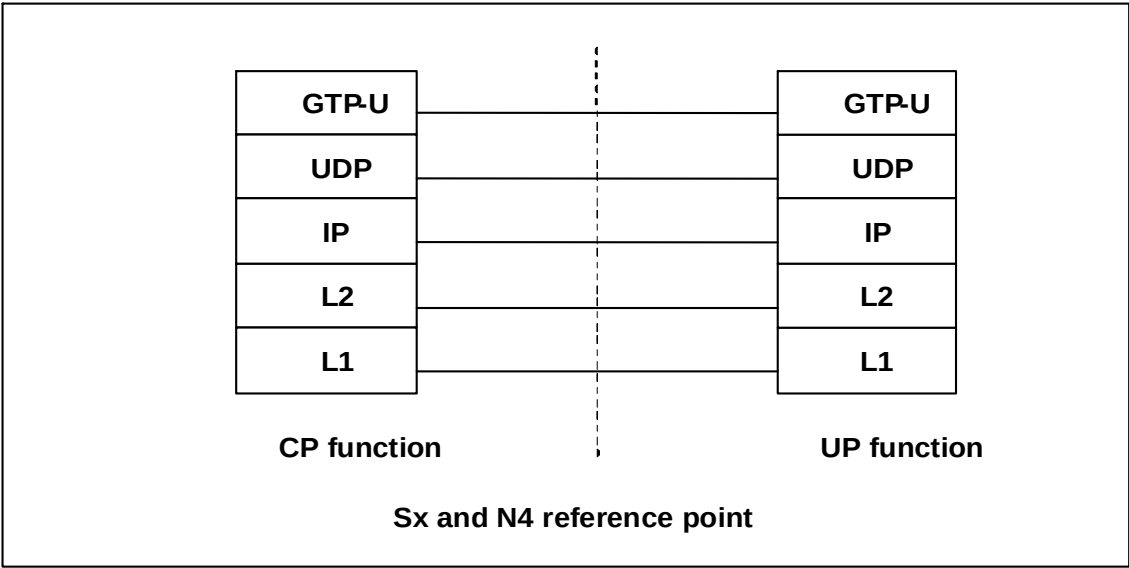


Figure 4.1-2: User Plane stack over Sxa, Sxb, combined Sxa/Sxb and N4

4.2 UDP Header and Port Numbers

4.2.1 General

A User Datagram Protocol (UDP) compliant with IETF RFC 768 [4] shall be used.

4.2.2 Request Message

The UDP Destination Port number for a Request message shall be 8805. It is the registered port number for PFCP.

The UDP Source Port for a Request message is a locally allocated port number at the sending entity.

NOTE: The locally allocated source port number can be reused for multiple Request messages.

4.2.3 Response Message

The UDP Destination Port value of a Response message shall be the value of the UDP Source Port of the corresponding Request message.

The UDP Source Port of a Response message shall be the value from the UDP Destination Port of the corresponding message.

4.3 IP Header and IP Addresses

4.3.1 General

In this clause, "IP" refers either to IPv4 as defined by IETF RFC 791 [5] or IPv6 as defined by IETF RFC 2460 [6]. A PFCP entity shall support both IPv4 and IPv6.

4.3.2 Request Message

The IP Destination Address of a Request message shall be an IP address of the peer entity.

5.4.12 Service Identification for Improved Radio Utilisation for GERAN

Service Identification for improved radio utilization for GERAN refers to the process in the PGW of marking DL user plane traffic with a Service Class Indicator (SCI) value. See clause 5.3.5.3 of 3GPP TS 23.060 [19].

This is controlled by the PGW-C by associating a QER, including the Service Class Indicator within the DL Flow Level Marking IE, to the PDR matching the DL traffic to be marked. The PGW-U performs the SCI marking for the detected DL traffic and sends the packet with the GTP-U Service Class Indicator Extension Header downstreams.

The PGW-C may stop the SCI marking during the PFCP session by removing the related QER or removing the DL Flow Level Marking IE from the related QER, the PGW-U shall then stop such function consequently.

5.4.13 Transport Level Marking

For EPC, transport level marking is performed on a per EPS bearer basis in the SGW and PGW. Transport level marking refers to the process of marking traffic with a DSCP value based on the locally configured mapping from the QCI and optionally the ARP priority level.

For 5GC, transport level marking is performed on a per QoS flow basis. Transport level marking refers to the process of marking traffic at the UPF/MB-UPF with a DSCP value based on the mapping from the 5QI, the Priority Level (if explicitly signalled) and optionally the ARP priority level configured at the SMF/MB-SMF.

Transport level marking shall be controlled by the CP function by providing the DSCP in the ToS or Traffic Class within the Transport Level Marking IE in the FAR that is associated to the PDR matching the traffic to be marked. The UP function shall perform the transport level marking for the detected traffic and sends the marked packet to the peer entity.

The CP function may change transport level marking by changing the Transport Level Marking IE in the related FAR.

5.4.14 Deferred PDR activation and deactivation

As specified in clause 6.3.2 of 3GPP TS 23.203 [7] and clauses 4.5.13 and 4a.5.13 of 3GPP TS 29.212 [8], Policy and charging control rule operations can be also performed in a deferred mode. To support such deferred PCC rule activation or deactivation, the CP function and UP function may optionally support the Deferred PDR activation and deactivation (DPDRA) as described below.

If the feature DPDRA is supported in both CP function and UP function, and when a PCC rule is provisioned in a deferred mode, i.e. with a Rule-Activation-Time and/or Rule-Deactivation-Time, the CP function shall provision the corresponding PDR using Create PDR IE or Update PDR together with an Activation Time and/or Deactivation Time to enable the PDR being activated or deactivated at requested time.

Without being provisioned together with an Activation Time or a Deactivation Time, a PDR shall be active immediately when it is received. When the status of a PDR is changed from active to inactive, the UP function shall keep storing the inactive PDR together with its associated FAR, URR(s) and QER(s), and then UP function shall apply the same behavior as if the PDR is deleted.

The CP function shall control at what time the status of a PDR rule changes using Activation Time and/or Deactivation Time as exactly as being instructed by the PCRF using a Rule-Activation-Time and/or Rule-Deactivation-Time, as specified in clause 4.5.13 of 3GPP TS 29.212 [8].

- 1) If only Activation Time is specified and has not yet occurred, then the UP function shall set the PDR rule inactive and make it active at that time. If Activation Time has passed, then the UP function shall immediately set the PDR rule active.
- 2) If only Deactivation Time is specified and has not yet occurred, then the UP function shall set the PDR rule active and make it inactive at that time. If Deactivation Time has passed, then the UP function shall immediately set the PDR rule inactive.
- 3) If both Activation Time and Deactivation Time are specified, and the Activation Time occurs before the Deactivation Time, and also when the PDR rule is provided before or at the time specified in the Deactivation Time, the UP function shall handle the rule as defined in 1) and then as defined in 2).

- allocate resources in the UP function using the Network Instance and, if available, the S-NSSAI.

A UP function that does not support allocating UP resources based on the Network Instance IE and S-NSSAI IE or that is not configured to do so by operator policy shall allocate UP resources based on the Network Instance IE (and other N4 information).

If the UP function supports the UE IP Address Usage Reporting feature (see clause 5.21.3.2), the UP function shall report the UP IP address usage as specified in clause 5.21.3.2, by reporting the UE IP address usage:

- a) per Network Instance, if the UP function does not support allocating UP resources based on the Network Instance IE and S-NSSAI IE or it is not configured to do so by operator policy; or

NOTE 2: This allows to report UE IP address usage per network slice when using a distinct Network Instance value per network slice.

- b) per Network Instance and S-NSSAI, if the UP function supports allocating UP resources based on the Network Instance IE and S-NSSAI IE and is configured to do so by operator policy. In this case, the UP Function shall report the UE IP address usage information associated with a specific S-NSSAI, as specified in clause 5.21.3.

NOTE 3: If the UP function indicates that it supports the "Per Slice UP Resource Management" feature and if this feature is also supported by the CP function, the CP function can instruct the UP function to allocate UP resource taking into account the network slice by including the Network Instance IE and S-NSSAI IE in the PFCP session establishment procedure. Otherwise, a separate Network Instance needs to be configured for each S-NSSAI if support of multiple slices is needed.

5.36 Paging Policy Indicator Provisioning

5.36.1 General

Stage 2 requirements for Paging Policy Differentiation are specified in clause 5.4.3.2 of 3GPP TS 23.501 [28].

For a UE in RRC Inactive state the NG-RAN can be required to enforce specific paging policies in the case of NG-RAN paging, based on 5QI, ARP and PPI associated with an incoming DL PDU. To support this, the SMF shall instruct the UPF to detect the DSCP in the TOS (IPv4) / TC (IPv6) value in the IP header of the DL PDU and insert the corresponding PPI as described below. When Paging Policy Differentiation is enabled in the network and a PPI needs to be inserted into DL PDU, the SMF shall:

- provision a PDR for each PPI containing one or more DSCP code(s) to be matched against the DSCP in the TOS (IPv4) / TC (IPv6) value in the IP header of the DL PDUs, where these DSCP code(s) are to be mapped to the same PPI, in the PDI together with or without the service data flow detection information;

NOTE 1: When service data flow level packet detection is not required, e.g. for an I/V-UPF, the PDR can be provisioned with one or more DSCP(s) to be matched in the PDI without any service data flow detection information.

NOTE 2: When service data flow level packet detection is required, several PDRs with the same service data flow detection information but with different (set of) DSCP(s) (which are mapped to different PPI) need to be provisioned.

- associate the PDR with a QER with the Paging Policy Indicator set to the mapped PPI to instruct UPF to insert the corresponding PPI into the GTP-U PDU Session Container extension header of outgoing packets.

5.36.2 Enhanced Paging Policy Indicator Provisioning

To reduce the number of PDRs/QERs to be provisioned (in order to support NG-RAN paging using Paging Policy Indicator (PPI)), the SMF and the UPF may support the Enhanced Provisioning Paging Policy Indicator (EPPPI) feature as specified below.

During a PFCP session establishment/modification procedure, if both the SMF and the UPF support the EPPPI feature, the SMF may include a DSCP to PPI Control Information in the PFCP Session Establishment/Modification Request message to instruct the UPF to insert a PPI in the GTP-U PDU Session Container extension header of downlink GTP-U

- the Load Control Information, Overload Control Information and the Failed Rule ID IEs may be included in a PFCP session related message.

If the Cause IE at Grouped IE level contains a value that indicates that the Grouped IE is not handled correctly, the other IEs in this Grouped IE, other than the Cause IE, may not be included.

7.2.3.3 Grouped Information Elements

A Grouped IE is an IE which may contain other IEs.

Grouped IEs have a length value in the TLV encoding, which includes the added length of all the embedded IEs. Overall coding of a grouped IE with 4 octets long IE header is defined in clause 8.2. Each IE within a grouped IE also shall also contain 4 octets long IE header.

Grouped IEs are not marked by any flag or limited to a specific range of IE type values. The clause describing an IE in this specification shall explicitly state if it is a Grouped IE.

NOTE: Each entry into each Grouped IE creates a new scope level. Exit from the grouped IE closes the scope level. The PFCP message level is the top most scope.

If more than one grouped IEs of the same type, but for a different purpose are sent within the same message level, these IEs shall have different IE types.

If more than one grouped IEs of the same type and for the same purpose are sent within the same message level, these IEs shall have exactly the same IE type to represent a list.

Assigning the same IE type to grouped IEs which don't have the same content is not recommended, even if these grouped IEs are in different message levels.

7.2.3.4 Information Element Type

An IE in a PFCP message or Grouped IE is identified by its IE Type and described by a specific row in the corresponding tables in clause 7.

If several IEs with the same Type are included in a PFCP message or Grouped IE, they represent a list for the corresponding IE name.

An IE Type value uniquely identifies a specific IE.

One IE type value is specified for Vendor Specific IEs.

7.3 Message Types

The PFCP message types to be used over the Sxa, Sxb, Sxc and N4 reference points are defined in Table 7.3-1.

the Sxa, Sxb, Sxc and N4 reference points are defined in Table 7.3-1.

Table 7.3-1: Message Types

Message Type value (Decimal)	Message	Applicability				
		Sxa	Sxb	Sxc	N4	N4 mb

0	Reserved					
	PFCP Node related messages					
1	PFCP Heartbeat Request	X	X	X	X	X
2	PFCP Heartbeat Response	X	X	X	X	X
3	PFCP PFD Management Request	-	X	X	X	-
4	PFCP PFD Management Response	-	X	X	X	-
5	PFCP Association Setup Request	X	X	X	X	X
6	PFCP Association Setup Response	X	X	X	X	X
7	PFCP Association Update Request	X	X	X	X	X
8	PFCP Association Update Response	X	X	X	X	X
9	PFCP Association Release Request	X	X	X	X	X
10	PFCP Association Release Response	X	X	X	X	X
11	PFCP Version Not Supported Response	X	X	X	X	X
12	PFCP Node Report Request	X	X	X	X	X
13	PFCP Node Report Response	X	X	X	X	X
14	PFCP Session Set Deletion Request	X	X	-	X	-
15	PFCP Session Set Deletion Response	X	X	-	X	-
16	PFCP Session Set Modification Request	-	X	-	X	-
17	PFCP Session Set Modification Response	-	X	-	X	-
18 to 49	For future use					
	PFCP Session related messages					
50	PFCP Session Establishment Request	X	X	X	X	X
51	PFCP Session Establishment Response	X	X	X	X	X
52	PFCP Session Modification Request	X	X	X	X	X
53	PFCP Session Modification Response	X	X	X	X	X
54	PFCP Session Deletion Request	X	X	X	X	X
55	PFCP Session Deletion Response	X	X	X	X	X
56	PFCP Session Report Request	X	X	X	X	X
57	PFCP Session Report Response	X	X	X	X	X
58 to 99	For future use					
	Other messages					
100 to 255	For future use					

7.4 PFCP Node Related Messages

7.4.1 General

This clause specifies either node level or PFCP entity level messages used over the Sxa, Sxb, Sxc, N4 and N4mb reference points (see clause 6.2).

7.4.2 Heartbeat Messages

7.4.2.1 Heartbeat Request

7.4.2.1 Table 7.4.2.1-1: Information Elements in Heartbeat Request

Information elements	P	Condition / Comment	Appl.					IE Type
			Sxa	Sxb	Sxc	N4	N4mb	
Recovery Time Stamp	M	This IE shall contain the time stamp when the PFCP entity was started see clause 19A of 3GPP TS 23.007 [24].	X	X	X	X	X	Recovery Time Stamp
Source IP Address	O	This IE may be included when a Network Address Translation device is deployed in the network. See clause 19a in 3GPP TS 23.007 [24].	X	X	X	X	X	Source IP Address

MPTCP Parameters	C	This IE shall be present if the TCI flag in the MPTCP Control Information IE is set to "1" in the Request message and the UPF allocated resources for MPTCP.	-	-	-	X	-	MPTCP Parameters
ATSSS-LL Parameters	C	This IE shall be present if the LLI flag in ATSSS-LL Control Information IE is set to "1" in the Request message and the UPF allocated resources for ATSSS-LL.	-	-	-	X	-	ATSSS-LL Parameters
PMF Parameters	C	This IE shall be present if the PMFI flag in the PMF Control Information IE is set to "1" in the Request message and the UPF allocated resources for PMF. Several IEs with the same IE type may be present to provide multiple PMF Parameters.	-	-	-	X	-	PMF Parameters

The MPTCP Parameters grouped IE shall be encoded as shown in Figure 7.5.3.7-2.

Table 7.5.3.7-2: MPTCP Parameters IE within PFCP Session Establishment Response

Octet 1 and 2	MPTCP Parameters IE Type = 225 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
MPTCP Address Information	M	This IE shall carry the information of allocated MPTCP address.	-	-	-	X	-	MPTCP Address Information
UE Link-Specific IP Address	M	This IE shall carry the information of allocated UE link-specific IP address for MPTCP.	-	-	-	X	-	UE Link-Specific IP Address

The ATSSS-LL Parameters grouped IE shall be encoded as shown in Figure 7.5.3.7-3.

Table 7.5.3.7-3: ATSSS-LL Parameters IE within PFCP Session Establishment Response

Octet 1 and 2	ATSSS-LL Parameters IE Type = 226 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
ATSSS-LL Information	M	This IE shall indicate that resources have been allocated to the ATSSS functionality.	-	-	-	X	-	ATSSS-LL Information

The PMF Parameters grouped IE shall be encoded as shown in Figure 7.5.3.7-4.

Table 7.5.3.7-4: PMF Parameters IE within PFCP Session Establishment Response

Octet 1 and 2	PMF Parameters IE Type = 227 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
PMF Address Information	M	This IE shall contain the PMF Address Information.	-	-	-	X	-	PMF Address Information
QoS flow identifier	O	If present, this IE shall identify the QoS Flow Identifier of the QoS flow related to the PMF address information.	-	-	-	X	-	QFI

7.5.3.8 Void

7.5.4 PFCP Session Modification Request

7.5.4.1 General

The PFCP Session Modification Request is used over the Sxa, Sxb, Sxc, N4 and N4mb interface by the CP function to request the UP function to modify the PFCP session.

Table 7.5.4.1-1: Information Elements in a PFCP Session Modification Request

Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	

CP F-SEID	C	This IE shall be present if the CP function decides to change its F-SEID for the PFCP session. The UP function shall use the new CP F-SEID for subsequent PFCP Session related messages for this PFCP Session. See Note 2.	X	X	X	X	X	F-SEID
Remove PDR	C	When present, this IE shall contain the PDR Rule which is requested to be removed. See Table 7.5.4-6-1. Several IEs within the same IE type may be present to represent a list of PDRs to remove.	X	X	X	X	X	Remove PDR
Remove FAR	C	When present, this IE shall contain the FAR Rule which is requested to be removed. See Table 7.5.4-7-1. Several IEs within the same IE type may be present to represent a list of FARs to remove.	X	X	X	X	X	Remove FAR
Remove URR	C	When present, this shall contain the URR Rule which is requested to be removed. See Table 7.5.4-8-1. Several IEs within the same IE type may be present to represent a list of URRs to remove.	X	X	X	X	FF S	Remove URR
Remove QER	C	When present, this IE shall contain the QER Rule which is requested to be removed. See Table 7.5.4-9-1. Several IEs within the same IE type may be present to represent a list of QERs to remove.	-	X	X	X	X	Remove QER
Remove BAR	C	When present, this IE shall contain the BAR Rule which is requested to be removed. See Table 7.5.4.12-1.	X	-	-	X	-	Remove BAR
Remove Traffic Endpoint	C	When present, this IE shall contain the Traffic Endpoint ID identifying the traffic endpoint to be removed, if the UP function has indicated support of PDI optimization. All the PDRs that refer to the removed Traffic Endpoint shall be deleted. See Table 7.5.4.14-1. Several IEs within the same IE type may be present to represent a list of Traffic Endpoints to remove.	X	X	X	X	X	Remove Traffic Endpoint
Create PDR	C	This IE shall be present if the CP function requests the UP function to create a new PDR. See Table 7.5.2.2-1. Several IEs within the same IE type may be present to represent a list of PDRs to create.	X	X	X	X	X	Create PDR
Create FAR	C	This IE shall be present if the CP function requests the UP function to create a new FAR. See Table 7.5.2.3-1. Several IEs within the same IE type may be present to represent a list of FARs to create.	X	X	X	X	X	Create FAR
Create URR	C	This IE shall be present if the CP function requests the UP function to create a new URR. See Table 7.5.2.4-1. Several IEs within the same IE type may be present to represent a list of URRs to create.	X	X	X	X	FF S	Create URR
Create QER	C	This IE shall be present if the CP function requests the UP function to create a new QER. See Table 7.5.2.5-1. Several IEs within the same IE type may be present to represent a list of QERs to create.	-	X	X	X	X	Create QER
Create BAR	C	This IE shall be present if the CP function requests the UP function to create a new BAR. See Table 7.5.2.6-1.	X	-	-	X	-	Create BAR
Create Traffic Endpoint	C	When present this IE shall contain the information associated with the Traffic Endpoint to be created, if the UP function has indicated support of PDI optimization. See Table 7.5.2.7-1. Several IEs within the same IE type may be present to represent a list of Traffic Endpoints to create.	X	X	X	X	X	Create Traffic Endpoint
Update PDR	C	This IE shall be present if a PDR previously created for the PFCP session need to be modified. See Table 7.5.4.2-1. Several IEs within the same IE type may be present to represent a list of PDRs to update.	X	X	X	X	X	Update PDR
Update FAR	C	This IE shall be present if a FAR previously created for the PFCP session need to be modified. See Table 7.5.4.3-1. Several IEs within the same IE type may be present to represent a list of FARs to update.	X	X	X	X	X	Update FAR

Update URR	C	This IE shall be present if URR(s) previously created for the PFCP session need to be modified. Several IEs within the same IE type may be present to represent a list of modified URRs. Previously URRs that are not modified shall not be included. See Table 7.5.4.4-1.	X	X	X	X	FF S	Update URR
Update QER	C	This IE shall be present if QER(s) previously created for the PFCP session need to be modified. Several IEs within the same IE type may be present to represent a list of modified QERs. Previously created QERs that are not modified shall not be included. See Table 7.5.4.5-1.	-	X	X	X	X	Update QER
Update BAR	C	This IE shall be present if a BAR previously created for the PFCP session needs to be modified. A previously created BAR that is not modified shall not be included. See Table 7.5.4.11-1.	X	-	-	X	-	Update BAR
Update Traffic Endpoint	C	When present this IE shall contain the information associated with the traffic endpoint to be updated, if the UP function has indicated support of PDI optimization. All the PDRs that refer to the Traffic Endpoint shall use the updated Traffic Endpoint information. See Table 7.5.4.13-1. Several IEs within the same IE type may be present to represent a list of Traffic Endpoints to update.	X	X	X	X	X	Update Traffic Endpoint
PFCPSMReq-Flags	C	This IE shall be included if at least one of the flags is set to "1". - DROBU (Drop Buffered Packets): the CP function shall set this flag if the UP function is requested to drop the packets currently buffered for this PFCP session (see NOTE 1). - QAURR (Query All URRs): the CP function shall set this flag if the CP function requests immediate usage report(s) for all the URRs previously provisioned for this PFCP session (see NOTE 3). - SUMPC (Stop Usage Measurement to Pause Charging): the CP function, e.g. PGW-C or (H-)SMF, shall set this flag if the usage measurement for the URRs which are applicable for charging (i.e. with the "ASPOC" flag set to "1") shall be stopped in the UP function. - RUMUC (Resume Usage Measurement to Un-pause Charging): the CP function, e.g. PGW-C or (H-)SMF, shall set this flag if the usage measurement for the URRs which are applicable for charging (i.e. with the "ASPOC" flag set to "1") shall be resumed in the UP function.	X X - -	- X X	- X -	X X X	- FF S -	PFCPSMReq-Flags
Query URR	C	This IE shall be present if the CP function requests immediate usage report(s) to the UP function. Several IEs within the same IE type may be present to represent a list of URRs for which an immediate report is requested. See Table 7.5.4.10-1. See NOTE 3.	X	X	X	X	FF S	Query URR
PGW-C/SMF FQ-CSID	C	This IE shall be included according to the requirements in clause 23 of 3GPP TS 23.007 [24] and clause 4.6 of 3GPP TS 23.527 [40].	X	X	-	X	-	FQ-CSID
SGW-C FQ-CSID	C	This IE shall be included according to the requirements in clause 23 of 3GPP TS 23.007 [24].	X	X	-	-	-	FQ-CSID
MME FQ-CSID	C	This IE shall be included according to the requirements in clause 23 of 3GPP TS 23.007 [24].	X	X	-	-	-	FQ-CSID
ePDG FQ-CSID	C	This IE shall be included according to the requirements in clause 23 of 3GPP TS 23.007 [24].	-	X	-	-	-	FQ-CSID

TSN FQ-CSID	C	This IE shall be included according to the requirements in clause 23 of 3GPP TS 23.007 [24].	-	X	-	-	-	FQ-CSID
User Plane Inactivity Timer	C	This IE shall be present if it needs to be changed.	-	X	X	X	X	User Plane Inactivity Timer
Query URR Reference	O	This IE may be present if the Query URR IE is present or the QAURR flag is set to "1". When present, it shall contain a reference identifying the query request, which the UP function shall return in any usage report sent in response to the query.	X	X	X	X	FF S	Query URR Reference
Trace Information	O	When present, this IE shall contain the trace instructions to be applied by the UP function for this PFCP session. A Trace Information with a null length indicates that the trace session shall be deactivated.	X	X	X	X	-	Trace Information
Remove MAR	C	When present, this IE shall contain the MAR Rule which is requested to be removed. See Table 7.5.4.15-1. Several IEs within the same IE type may be present to represent a list of MARs to remove.	-	-	-	X	-	Remove MAR
Update MAR	C	This IE shall be present if an MAR previously created for the PFCP session needs to be modified. See Table 7.5.4.16-1. Several IEs within the same IE type may be present to represent a list of MARs to update.	-	-	-	X	-	Update MAR
Create MAR	C	This IE shall be present if the CP function requests the UP function to create a new MAR for a new PDR. See Table 7.5.2.8-1. Several IEs within the same IE type may be present to represent a list of MARs to create.	-	-	-	X	-	Create MAR
Node ID	C	This IE shall be present if a new SMF in an SMF Set, with one PFCP association per SMF and UPF (see clause 5.22.3), takes over the control of the PFCP session. When present, it shall contain the unique identifier of the new SMF.	-	-	-	X	X	Node ID
TSC Management Information	C	This IE shall be present if the SMF needs to send TSC Management information to the UPF. Several IEs within the same IE type may be present to transfer PMICs for different NW-TT ports.	-	-	-	X	-	TSC Management Information
Remove SRR	C	When present, this shall indicate the SRR Rule which is requested to be removed. See Table 7.5.4.19-1. Several IEs within the same IE type may be present to represent a list of SRRs to remove.	-	-	-	X	-	Remove SRR
Create SRR	C	This IE shall be present if the CP function requests the UP function to create a new SRR. See Table 7.5.2.9-1. Several IEs within the same IE type may be present to represent a list of SRRs to create.	-	-	-	X	-	Create SRR
Update SRR	C	This IE shall be present if SRR(s) previously created for the PFCP session need to be modified. Several IEs within the same IE type may be present to represent a list of modified SRRs. Previously SRRs that are not modified shall not be included. See Table 7.5.4.20-1.	-	-	-	X	-	Update SRR
Provide ATSSS Control Information	C	This IE shall be present for PFCP session modification for an MA PDU session, if the ATSSS Control Information changes. When present, this IE shall contain the required ATSSS functionalities for this MA PDU session. The UPF shall replace any value received previously by the new information received in this IE. See Note 4. See Table 7.5.2.10-1.	-	-	-	X	-	Provide ATSSS Control Information
Ethernet Context Information	C	This IE shall be present to update the list of MAC addresses associated to the PDU session during an Ethernet PDU session anchor relocation.	-	-	-	X	-	Ethernet Context Information

Access Availability Information	O	This IE may be present for an MA PDU session to signal that an access type has become transiently unavailable or has become available again (see clause 5.20.5). Two IEs with the same IE type may be present to report changes of access availability for both 3GPP and non-3GPP accesses.	-	-	-	X	-	Access Availability Information
Query Packet Rate Status	C	This IE shall be present if the CP function requests immediate packet rate status report(s) to the UP function. Several IEs within the same IE type may be present to represent a list of QERs for which an immediate packet rate status report is requested. See Table 7.5.4.22-1.	-	X	-	X	-	Query Packet Rate Status
S-NSSAI	O	This IE may be present to indicate the S-NSSAI of the PDU session or MBS session, if the S-NSSAI of the PDU Session or MBS session has been provided previously to the UP function and the S-NSSAI has changed. (NOTE 5)	-	-	-	X	X	S-NSSAI
RAT Type		This IE shall be present if there is a RAT change and the CP function wants to inform the UP Function about the new RAT Type if the PFCP session is not established for a MA PDU session.	X	X	-	X	-	RAT Type
Group Id	C	This IE shall be included by the CP function to indicate the new group identifier to which the PFCP session pertains, if it has been changed (see clause 5.22.4). When present, the UP function shall replace any earlier value associated to the PFCP session with the new value.	-	X	-	X	-	Group Id
MBS Session N4 Control Information	C	This IE shall be included if: - the correspond PDU session is being associated with an MBS session, or with an MBS session and Area Session ID for a location dependent MBS service; or - the PDU session is already associated with an MBS session, or with an MBS session and Area Session ID for a location dependent MBS service, and the N19mb multicast transport information has changed. In this case, the Multicast Transport Information IE shall contain the new multicast transport information to use to receive MBS session data from the MB-UPF. Several IEs with the same IE type may be present to provide N4 control information for several MBS sessions, e.g. when the UE requests to join several MBS sessions. See Table 7.5.2.1-5 for encoding.	-	-	-	X	-	MBS Session N4 Control Information
DSCP to PPI Control Information	C	This IE shall be present if the DSCP to PPI Control Information needs to be changed. The UPF shall replace any value received previously by the new information received in this IE. See Table 7.5.2.1-6 for encoding.	-	-	-	X	-	DSCP to PPI Control Information

NOTE 1:	The CP function may request the UP function to drop the packets currently buffered for the PFCP session when using extended buffering of downlink data packets, buffering is performed in the UP function and the DL Data Buffer Expiration Time is handled by the CP function. In this case, when the DL Data Buffer Expiration Time expires, the CP function shall send a PFCP Session Modification Request including the DROBU flag (to drop the downlink data packets currently buffered in the UP function) and updating the Apply Action within the FARs of this PFCP session to request the UP function to start buffering the downlink data packets with notifying the arrival of subsequent downlink data packets. See clause 5.9.3 of 3GPP TS 23.214 [2].
NOTE 2:	When changing the CP F-SEID of an established PFCP Session, the CP function shall be able to handle any incoming PFCP Session related messages sent by the UP function with the previous CP F-SEID for a duration at least longer than twice the PFCP retransmission timer (N1xT1).
NOTE 3:	The QAURR (Query All URRs) flag in the PFCPSMReq-Flags IE and the Query URR IE are exclusive from each other in a PFCP Session Modification Request.
NOTE 4:	If the ATSSS resources have already been allocated to the PFCP session previously, e.g. during the PFCP session establishment, the UPF shall not allocate new values for such resources (e.g. UE Link-Specific IP Address).
NOTE 5:	S-NSSAI for the PDU session may be updated after PDU session establishment, i.e. during EPS to 5GS handover procedure, the initial AMF may use configured S-NSSAI for interworking to create the PDU session in 5GS. For home routed PDU session, if the S-NSSAI in serving PLMN (mapped from S-NSSAI in HPLMN) is different from the configured S-NSSAI for interworking and V-SMF reselection is not needed, the AMF will update V-SMF with S-NSSAI in serving PLMN for the PDU session, as specified in clause 4.11.1.3.3 of 3GPP TS 23.502 [29]. The S-NSSAI may also be updated by SMF+PGW-C during EPS to 5GS handover in non-roaming or roaming-with-LBO case. The S-NSSAI may be used by the UP function for performance measurement.

7.5.4.2 Update PDR IE within PFCP Session Modification Request

The Update PDR grouped IE shall be encoded as shown in Figure 7.5.4.2-1.

Table 7.5.4.2-1: Update PDR IE within PFCP Session Modification Request

Octet 1 and 2	Update PDR IE Type = 9 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	

PDR ID	M	This IE shall uniquely identify the PDR among all the PDRs configured for that PFCP session.	X	X	X	X	X	PDR ID
Outer Header Removal	C	This IE shall be present if it needs to be changed.	X	X	-	X	X	Outer Header Removal
Precedence	C	This IE shall be present if there is a change in the PDR's precedence to be applied by the UP function among all PDRs of the PFCP session, when looking for a PDR matching an incoming packet.	-	X	X	X	X	Precedence
PDI	C	This IE shall be present if there is a change within the PDI against which incoming packets will be matched. When present, this IE shall replace the PDI previously stored in the UP function for this PDR. See Table 7.5.2.2-2.	X	X	X	X	X	PDI
FAR ID	C	This IE shall be present if it needs to be changed	X	X	X	X	X	FAR ID
URR ID	C	This IE shall be present if a measurement action shall be applied or no longer applied to packets matching this PDR. When present, this IE shall contain the list of all the URR IDs to be associated to the PDR.	X	X	X	X	FF S	URR ID
QER ID	C	This IE shall be present if a QoS enforcement action shall be applied or no longer applied to packets matching this PDR. When present, this IE shall contain the list of all the QER IDs to be associated to the PDR.	-	X	X	X	X	QER ID
Activate Predefined Rules	C	This IE shall be present if new Predefined Rule(s) needs to be activated for the PDR. When present this IE shall contain one Predefined Rules name. Several IEs with the same IE type may be present to represent multiple "Activate Predefined Rules" names.	-	X	X	X	-	Activate Predefined Rules
Deactivate Predefined Rules	C	This IE shall be present if Predefined Rule(s) needs to be deactivated for the PDR. When present this IE shall contain one Predefined Rules name. Several IEs with the same IE type may be present to represent multiple "Activate Predefined Rules" names.	-	X	X	X	-	Deactivate Predefined Rules
Activation Time	O	This IE may be present if the PDR activation time shall be changed. (NOTE 2)	-	X	X	X	-	Activation Time
Deactivation Time	O	This IE may be present if the PDR deactivation time shall be changed. (NOTE 2)	-	X	X	X	-	Deactivation Time
IP Multicast Addressing Info	O	This IE may be present in an UL PDR controlling UL IGMP/MLD traffic (see clause 5.25), if it needs to be changed When present, it shall contain a (range of) IP multicast address(es), and optionally source specific address(es), identifying a set of IP multicast flows. See Table 7.5.2.2-4. Several IEs with the same IE type may be present to represent multiple IP multicast flows. When present, the UPF shall replace any IP multicast address(es) previously stored for this PDR by the IP multicast address(es) received in this IE.	-	-	-	X	-	IP Multicast Addressing Info
Transport Delay Reporting	C	This IE shall be present if Transport Delay Reporting needs to be changed (e.g. transport delay reporting needs to be activated or deactivated). See Table 7.5.2.2-6.	-	-	-	X	-	Transport Delay Reporting
RAT Type	O	This IE may be present if there is a RAT change for the UL PDR if the PFCP session is established for a MA PDU session.	-	-	-	X	-	RAT Type
NOTE1: The IEs which do not need to be modified shall not be included in the Update PDR IE. The UP function shall continue to behave according to the values previously received for IEs not present in the Update PDR IE.								
NOTE2: When the Activation Time and Deactivation Time are not present, the PDR shall keep its current activation status, either active or inactive.								

7.5.4.3 Update FAR IE within PFCP Session Modification Request

The Update FAR grouped IE shall be encoded as shown in Figure 7.5.4.3-1.

Table 7.5.4.3-1: Update FAR IE within PFCP Session Modification Request

Octet 1 and 2		Update FAR IE Type = 10 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sxa	Sxb	Sxc	N4	N4mb	
FAR ID	M	This IE shall identify the FAR to be updated.	X	X	X	X	X	FAR ID
Apply Action	C	This IE shall be present if it is changed.	X	X	X	X	X	Apply Action
Update Forwarding parameters	C	This IE shall be present if it is changed. See table 7.5.4.3-2.	X	X	X	X	-	Update Forwarding Parameters
Update Duplicating Parameters	C	This IE shall be present if it is changed. See table 7.5.4.3-3. Several IEs with the same IE type may be present to request to duplicate the packets to different destinations.	X	X	-	-	-	Update Duplicating Parameters
Redundant Transmission Forwarding Parameters	C	This IE shall be present if it is changed. See table 7.5.2.3-4.	-	-	-	X	-	Redundant Transmission Forwarding Parameters
BAR ID	C	This IE shall be present if the BAR ID associated to the FAR needs to be modified.	X	-	-	X	-	BAR ID
Add MBS Unicast Parameters	C	This IE shall be present when the Apply Action is set to "MBSU" to request the MB-UPF to forward the MBS session data to a remote GTP-U peer for unicast transport. Several IEs with the same IE type may be present to represent multiple remote GTP-U peers. See Table 7.5.2.3-6.	-	-	-	-	X	Add MBS Unicast Parameters
Remove MBS Unicast Parameters	C	This IE shall be present when the MB-SMF to request the MB-UPF to delete a remote GTP-U peer to stop forwarding the MBS session data to that remote GTP-U peer for unicast transport. Several IEs with the same IE type may be present to represent to delete multiple remote GTP-U peers.	-	-	-	-	X	Remove MBS Unicast Parameters

Table 7.5.4.3-2: Update Forwarding Parameters IE in the Update FAR IE

Octet 1 and 2		Update Forwarding Parameters IE Type = 11 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sxa	Sxb	Sxc	N4	N4mb	

Destination Interface	C	This IE shall only be provided if it is changed. When present, it shall indicate the destination interface of the outgoing packet.	X	X	X	X	-	Destination Interface
Network instance	C	This IE shall only be provided if it is changed.	X	X	X	X	-	Network Instance
Redirect Information	C	This IE shall be present if the instructions regarding the redirection of traffic by the UP function need to be modified.	-	X	X	X	-	Redirect Information
Outer Header Creation	C	This IE shall only be provided if it is changed. See NOTE 1.	X	X	-	X	-	Outer Header Creation
Transport Level Marking	C	This IE shall only be provided if it is changed	X	X	-	X	-	Transport Level Marking
Forwarding Policy	C	This IE shall only be provided if it is changed. See NOTE 1.	-	X	X	X	-	Forwarding Policy
Header Enrichment	C	This IE shall only be provided if it is changed	-	X	X	X	-	Header Enrichment
PFCPSMReq-Flags	C	This IE shall be included if at least one of the flags is set to "1". - SNDEM (Send End Marker Packets): this IE shall be present if the CP function modifies the F-TEID of the downstream node in the Outer Header Creation IE and the CP function requests the UP function to construct and send GTP-U End Marker messages towards the old F-TEID of the downstream node.	X	X	-	X	-	PFCPSMReq-Flags
Linked Traffic Endpoint ID	C	This IE may be present, if it is changed and the UP function indicated support of the PDI optimization feature, (see clause 8.2.25). When present, it shall identify the Traffic Endpoint ID allocated for this PFCP session to receive the traffic in the reverse direction (see clause 5.2.3.1).	X	X	-	X	-	Traffic Endpoint ID
Destination Interface Type	C	This IE shall be present to indicate the 3GPP interface type of the destination interface, if the value has changed.	X	X	-	X	-	3GPP Interface Type
Data Network Access Identifier	C	This IE shall be provided over N16a if it is changed. This IE shall not be sent over N4.	-	-	-	-	-	Data Network Access Identifier
IP Address and Port Number Replacement	C	This IE shall only be provided if it is changed.	-	-	-	X	-	IP Address and Port Number Replacement
NOTE 1: If the Outer Header Creation and Forwarding Policy are present, the UP function shall put the user plane packets in the user plane tunnel by applying Outer Header Creation, after enforcing the required Forwarding Policy.								

Table 7.5.4.3-3: Update Duplicating Parameters IE in the Update FAR IE

Octet 1 and 2		Update Duplicating Parameters IE Type = 105 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
Destination Interface	C	This IE shall only be provided if it is changed. When present, it shall indicate the destination interface of the outgoing packet.	X	X	-	-	-	Destination Interface
Outer Header Creation	C	This IE shall only be provided if it is changed. See NOTE 1.	X	X	-	-	-	Outer Header Creation
Transport Level Marking	C	This IE shall only be provided if it is changed.	X	X	-	-	-	Transport Level Marking
Forwarding Policy	C	This IE shall only be provided if it is changed. See NOTE 1.	-	X	-	-	-	Forwarding Policy
NOTE 1: If the Outer Header Creation and Forwarding Policy are present, the UP function shall put the user plane packets in the user plane tunnel by applying Outer Header Creation, after enforcing the required Forwarding Policy.								

Table 7.5.4.3-4: Remove MBS Unicast Parameters IE in the Update FAR IE

Octet 1 and 2	Remove MBS Unicast Parameters Type = 304 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sxa	Sxb	Sxc	N4	N4 mb	
MBS Unicast Parameters ID	M	This IE shall identify the MBS Unicast Parameters IE to be removed.	-	-	-	-	X	MBS Unicast Parameters ID

7.5.4.4 Update URR IE within PCF Session Modification Request

The Update URR grouped IE shall be encoded as shown in Figure 7.5.4.4-1.

Table 7.5.4.4-1: Update URR IE within PCF Session Modification Request

Octet 1 and 2	Update URR IE Type = 13 (decimal)						
Octets 3 and 4	Length = n						
Information elements	P	Condition / Comment	Appl.				IE Type
			Sx a	Sx b	Sx c	N4	

URR ID	M	This IE shall uniquely identify the URR among all the URRs configured for that PFCP session	X	X	X	X	URR ID
Measurement Method	C	This IE shall be present if the measurement method needs to be modified. When present, this IE shall indicate the method for measuring the network resources usage, i.e. whether the data volume, duration (i.e. time), combined volume/duration, or event shall be measured.	X	X	X	X	Measurement Method
Reporting Triggers	C	This IE shall be present if the reporting triggers needs to be modified. When present, this IE shall indicate the trigger(s) for reporting network resources usage to the CP function, e.g. periodic reporting or reporting upon reaching a threshold, or envelope closure, or when an SMF instructs an UPF to report the reception of the End Marker packet from the old I-UPF during a Service Request procedure (see clauses 4.2.3.2 and 4.23.4.3 in 3GPP TS 23.502 [29]).	X	X	X	X	Reporting Triggers
Measurement Period	C	This IE shall be present if the Measurement Period needs to be modified. When present, it shall indicate the period for generating and reporting usage reports.	X	X	X	X	Measurement Period
Volume Threshold	C	This IE shall be present if the Volume Threshold needs to be modified. When present, it shall indicate the traffic volume value after which the UP function shall report network resources usage to the CP function for this URR.	X	X	X	X	Volume Threshold
Volume Quota	C	This IE shall be present if the Volume Quota needs to be modified. When present, it shall indicate the Volume Quota value.	-	X	X	X	Volume Quota
Time Threshold	C	This IE shall be present if the Time Threshold needs to be modified. When present, it shall indicate the time usage after which the UP function shall report network resources usage to the CP function for this URR.	X	X	X	X	Time Threshold
Time Quota	C	This IE shall be present if the Time Quota needs to be modified. When present, it shall indicate the Time Quota value.	-	X	X	X	Time Quota
Event Threshold	C	This IE shall be present if Event Threshold needs to be modified. When present, it shall indicate the number of events after which the UP function shall report to the CP function for this URR.	-	X	X	X	Event Threshold
Event Quota	C	This IE shall be present if Event Quota needs to be modified. When present, it shall indicate the Event Quota value.	-	X	X	X	Event Quota
Quota Holding Time	C	This IE shall be present if the Quota Holding Time needs to be modified. When present, it shall contain the duration of the Quota Holding Time.	-	X	X	X	Quota Holding Time
Dropped DL Traffic Threshold	C	This IE shall be present if the Dropped DL Threshold needs to be modified. When present, it shall contain the threshold of the DL traffic being dropped.	X	-	-	X	Dropped DL Traffic Threshold
Quota Validity Time	C	This IE shall be present if Quota Validity time was not sent earlier or quota validity time value needs to be modified.	-	X	-	X	Quota Validity Time
Monitoring Time	C	This IE shall be present if the Monitoring Time needs to be modified. When present, this IE shall contain the time at which the UP function shall re-apply the volume or time threshold.	X	X	X	X	Monitoring Time
Subsequent Volume Threshold	C	This IE shall be present if the Subsequent Volume Threshold needs to be modified and volume-based measurement is used. When present, it shall indicate the traffic volume value after which the UP function shall report network resources usage to the CP function for this URR for the period after the Monitoring Time.	X	X	X	X	Subsequent Volume Threshold

Subsequent Time Threshold	C	This IE shall be present if the Subsequent Time Threshold needs to be modified. When present, it shall indicate the time usage value after which the UP function shall report network resources usage to the CP function for this URR for the period after the Monitoring Time.	X	X	X	X	Subsequent Time Threshold
Subsequent Volume Quota	C	This IE shall be present if the Subsequent Volume Quota needs to be modified. When present, it shall indicate the Volume Quota value which the UP function shall use for this URR for the period after the Monitoring Time.	-	X	X	X	Subsequent Volume Quota
Subsequent Time Quota	C	This IE shall be present if the Subsequent Time Quota needs to be modified. When present, it shall indicate the Time Quota value which the UP function shall use for this URR for the period after the Monitoring Time.	-	X	X	X	Subsequent Time Quota
Subsequent Event Threshold	O	This IE shall be present if the Subsequent Event Threshold needs to be modified. When present, it shall indicate the number of events after which the UP function shall report to the CP function for this URR for the period after the Monitoring Time.	-	X	X	X	Subsequent Event Threshold
Subsequent Event Quota	O	This IE shall be present if the Subsequent Event Quota needs to be modified. When present, it shall indicate the Event Quota value which the UP function shall use for this URR for the period after the Monitoring Time.	-	X	X	X	Subsequent Event Quota
Inactivity Detection Time	C	This IE shall be present if the Inactivity Detection Time needs to be modified. When present, it shall indicate the duration of the inactivity period after which time measurement needs to be suspended when no packets are received during this inactivity period.	-	X	X	X	Inactivity Detection Time
Linked URR ID	C	This IE shall be present if linked usage reporting is required. When present, this IE shall contain the linked URR ID which is related with this URR (see clause 5.2.2.4). Several IEs with the same IE type may be present to represent multiple linked URRs which are related with this URR.	-	X	X	X	Linked URR ID

Measurement Information	C	his IE shall be included if any of the following flag is set to "1" or if the change of flag(s) from "1" to "0" results in the IE becoming set to all zeros. Applicable flags are: - Inactive Measurement Flag: this flag shall be set to "1" if the measurement shall be paused (inactive). The measurement shall be performed (active) if the bit is set to "0" or if the Measurement Information IE is not present in the Update URR IE. - Reduced Application Detection Information Flag: this flag may be set to "1", if the Reporting Triggers request to report the start or stop of application, to request the UP function to only report the Application ID in the Application Detection Information, e.g. for envelope reporting. - Immediate Start Time Metering Flag: this flag may be set to "1" if time-based measurement is used and the UP function is requested to start the time metering immediately at receiving the flag. - Send Start Pause of Charging Flag: this flag may be set to "1" by the CP function if the UP Function is requested to send a Start Pause of Charging indication to the upstream GTP-U entity(s) when the Dropped DL Traffic Threshold is reached. - Applicable for Start of Pause of Charging Flag: this flag may be set to "1" if the URR is applicable for Start of Pause of Charging, so that the UP function shall stop the usage measurement for the URR when receiving Start Pause of Charging indication from the peer downstream GTP-U entity. - Control of Inactive Measurement Flag: the flag shall be set to "1" if the CP function requests the UP function to stop or resume the usage measurement for the URR with the "ASPOC" flag set to "1" according to the value of Inactive Measurement Flag.	-	X	-	X	Measurement Information
Time Quota Mechanism	C	This IE shall be present if time-based measurement based on CTP or DTP needs to be modified.	-	X	-	-	Time Quota Mechanism
Aggregated URRs	C	This IE shall be included if the Aggregated URRs IE needs to be modified. See Table 7.5.2.4-2. Several IEs with the same IE type may be present to provision multiple aggregated URRs. When present, this IE shall provide the complete list of the aggregated URRs.	-	X	-	-	Aggregated URRs
FAR ID for Quota Action	C	This IE shall be present if the FAR ID for Quota Action IE needs to be modified. This IE may be present if the Volume Quota IE or the Time Quota IE or Event Quota IE is newly provisioned in the URR and the UP Function indicated support of the Quota Action. When present, it shall contain the identifier of the substitute FAR the UP function shall apply, for the traffic associated to this URR, when exhausting any of these quotas. See NOTE 1, NOTE 2.	-	X	X	X	FAR ID
Ethernet Inactivity Timer	C	This IE shall be present if the Ethernet Inactivity Timer needs to be modified. When present, it shall contain the duration of the Ethernet inactivity period.	-	-	-	X	Ethernet Inactivity Timer

Additional Monitoring Time	O	This IE shall be present if the additional Monitoring Time needs to be modified. When present, this IE shall contain the time at which the UP function shall re-apply the volume or time or event threshold/quota. See Table 7.5.2.4-3. The CP function shall provide the full set of Additional Monitoring Times IE(s). The UP function shall replace any Additional Monitoring Times IE(s) provisioned earlier by the new set of received IE(s).	X	X	X	X	Additional Monitoring Time
Number of Reports	O	This IE may be present if the Number of Reports need to be changed. When present, it shall indicate the number of usage reports to be generated by the URR. See also clauses 5.2.2.2.1 and 5.2.2.3.1.	X	X	X	X	Number of Reports
Exempted Application ID for Quota Action	C	This IE shall be present if Application ID for Quota Action needs to be changed. When present, it shall contain an Application ID matching packets that shall be exempted from applying the FAR ID for Quota Action when the quota has been exhausted. Several IEs with the same IE type may be present to provide multiple Application IDs. The CP function shall always provide a complete list of Application IDs. See NOTE 3.	-	X	X	X	Application ID
Exempted SDF Filter for Quota Action	C	This IE shall be present if the SDF Filter for Quota Action needs to be changed. When present, it shall contain a SDF Filter matching packets that shall be exempted from applying the FAR ID for Quota Action when the quota has been exhausted. Several IEs with the same IE type may be present to provide multiple SDF Filters. The CP function shall always provide a complete list of SDF Filters. See NOTE 3.	-	X	X	X	SDF Filter
User Plane Inactivity Timer	C	This IE shall be present if the User Plane Inactivity Timer needs to be modified. When present, it shall contain the duration of the User Plane Inactivity Timer.	-	-	-	X	User Plane Inactivity Timer
NOTE 1: The substitute FAR used when exhausting a Volume Quota or Time Quota may be set to drop the packets or redirect the traffic towards a redirect destination as specified in clause 5.4.7. NOTE 2: If the FAR as indicated in the FAR ID for Quota Action is removed after being provisioned, the UP function shall behave as if the FAR ID for Quota Action is not provisioned and shall apply the default behaviour per local configuration when the quota is exhausted. NOTE 3: The Exempted Application ID for Quota Action IE or Exempted Filter ID for Quota Action IE may be provisioned as the Restricted-Filter-Rule AVP or Filter ID AVP which is included in Final-Unit-Indication AVP from the online charging system when the Final-Unit-Action AVP is set to "REDIRECT" or "RESTRICT_ACCESS". See also 3GPP TS 32.299 [18].							

7.5.4.5 Update QER IE within PFCP Session Modification Request

The Update QER grouped IE shall be encoded as shown in Figure 7.5.4.5-1.

Table 7.5.4.5-1: Update QER IE within PFCP Session Modification Request

Octet 1 and 2	Update QER IE Type = 14 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	

QER ID	M	This IE shall uniquely identify the QER among all the QRS configured for that PFCP session	-	X	X	X	X	QER ID
QER Correlation ID	C	This IE shall be present if the QER correlation ID in this QER needs to be modified. See NOTE 1.	-	X	-	X	-	QER Correlation ID
Gate Status	C	This IE shall be present if the Gate Status needs to be modified. When present, it shall indicate whether the packets are allowed to be forwarded (the gate is open) or shall be discarded (the gate is closed) in the uplink and/or downlink directions. See NOTE 1.	-	X	X	X	X	Gate Status
Maximum Bitrate	C	This IE shall be present if an MBR enforcement action applied to packets matching this PDR need to be modified. When present, this IE shall indicate the uplink and/or downlink maximum bit rate to be enforced for packets matching the PDR. For EPC, this IE may be set to the value of: - the APN-AMBR, for a QER that is referenced by all the PDRs of the non-GBR bearers of a PDN connection; - the TDF session MBR, for a QER that is referenced by all the PDRs of a TDF session; - the bearer MBR, for a QER that is referenced by all the PDRs of a bearer; - the SDF MBR, for a QER that is referenced by all the PDRs of a SDF. For 5GC, this IE may be set to the value of: - the Session-AMBR, for a QER that is referenced by all the PDRs of the non-GBR QoS flows of a PDU session; - the QoS Flow MBR, for a QER that is referenced by all the PDRs of a QoS Flow; - the SDF MBR, for a QER that is referenced by all the PDRs of a SDF. See NOTE 1.	-	X	X	X	X	MBR
Guaranteed Bitrate	C	This IE shall be present if a GBR authorization to packets matching this PDR needs to be modified. When present, this IE shall indicate the authorized uplink and/or downlink guaranteed bit rate. This IE may be set to the value of: - the aggregate GBR, for a QER that is referenced by all the PDRs of a GBR bearer; - the QoS Flow GBR, for a QER that is referenced by all the PDRs of a QoS Flow (for 5GC); - the SDF GBR, for a QER that is referenced by all the PDRs of a SDF. See NOTE 1.	-	X	X	X	X	GBR
Packet Rate	C	This IE shall be present if a Packet Rate enforcement action (in terms of number of packets per time interval) need to be modified for packets matching this PDR.	-	X	-	X	-	Packet Rate
DL Flow Level Marking	C	This IE shall be set if the DL Flow Level Marking IE needs to be modified. See NOTE 1.	-	X	X	-	-	DL Flow Level Marking
QoS flow identifier	C	This IE shall be present if it needs to be modified.	-	-	-	X	X	QFI
Reflective QoS	C	This IE shall be present if the state of the Reflective QoS needs to be modified (activate if inactive, or deactivate the active Reflective QoS).	-	-	-	X	-	RQI
Paging Policy Indicator	C	This IE shall be present if it needs to be modified.	-	-	-	X	-	Paging Policy Indicator
Averaging Window	O	This IE may be present if the UP function is required to modify the Averaging Window. (NOTE 2)	-	-	-	X	-	Averaging Window

QER Control Indications	C	This IE shall be included if the CP function need to provide the updated QoS enforcement control information: - RCSR (Rate Control Status Reporting): the CP function shall set this bit "1" to request the UP function to report the rate control status when the PFCP session is released.	-	X	-	X	-	QER Control Indications
NOTE 1: The IEs which do not need to be modified shall not be included in the Update QER IE. The UP function shall continue to behave according to the values previously received for IEs not present in the Update QER IE.								
NOTE 2: As 5QI is not signalled over N4, one default averaging window shall be pre-configured in the UPF.								

7.5.4.6 Remove PDR IE within PFCP Session Modification Request

The Remove PDR grouped IE shall be encoded as shown in Figure 7.5.4.6-1.

R grouped IE shall be encoded as shown in Figure 7.5.4.6-1.

Table 7.5.4.6-1: Remove PDR IE within PFCP Session Modification Request

Octet 1 and 2	Remove PDR IE Type = 15 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
PDR ID	M	This IE shall identify the PDR to be deleted.	X	X	X	X	X	PDR ID

7.5.4.7 Remove FAR IE within PFCP Session Modification Request

The Remove FAR grouped IE shall be encoded as shown in Figure 7.5.4.7-1.

Table 7.5.4.7-1: Remove FAR IE within PFCP Session Modification Request

Octet 1 and 2	Remove FAR IE Type = 16 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
FAR ID	M	This IE shall identify the FAR to be deleted.	X	X	X	X	X	FAR ID

7.5.4.8 Remove URR IE within PFCP Session Modification Request

The Remove URR grouped IE shall be encoded as shown in Figure 7.5.4.7-1.

Table 7.5.4.8-1: Remove URR IE within PFCP Session Modification Request

Octet 1 and 2	Remove URR IE Type = 17 (decimal)						
Octets 3 and 4	Length = n						
Information elements	P	Condition / Comment	Appl.				IE Type
			Sx a	Sx b	Sx c	N4	
URR ID	M	This IE shall identify the URR to be deleted.	X	X	X	X	URR ID

7.5.4.9 Remove QER IE PFCP Session Modification Request

The Remove QER grouped IE shall be encoded as shown in Figure 7.5.4.9-1.

T Table 7.5.4.9-1: Remove QER IE PFCP Session Modification Request

Octet 1 and 2	Remove QER IE Type = 18 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
QER ID	M	This IE shall identify the QER to be deleted.	-	X	X	X	X	QER ID

7.5.4.10 Query URR IE within PFCP Session Modification Request

The Query URR grouped IE shall be encoded as shown in Figure 7.5.4.10-1.

Table 7.5.4.10-1: Query URR IE within PFCP Session Modification Request

Octet 1 and 2	Query URR IE Type = 77 (decimal)						
Octets 3 and 4	Length = n						
Information elements	P	Condition / Comment	Appl.				IE Type
			Sx a	Sx b	Sx c	N4	
URR ID	M	This IE shall identify the URR being queried.	X	X	X	X	URR ID

7.5.4.11 Update BAR IE within PFCP Session Modification Request

The Update BAR grouped IE shall be encoded as shown in Figure 7.5.4.11-1.

Table 7.5.4.11-1: Update BAR IE within PFCP Session Modification Request

Octet 1 and 2	Update BAR IE Type = 86 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
BAR ID	M	This IE shall identify the BAR Rule to be modified.	X	-	-	X	-	BAR ID
Downlink Data Notification Delay	C	This IE shall be present if the UP function indicated support of the Downlink Data Notification Delay parameter (see clause 8.2.28) and the Downlink Data Notification Delay needs to be modified. When present, it shall contain the delay the UP function shall apply between receiving a downlink data packet and notifying the CP function about it, when the Apply Action parameter requests to buffer the packets and notify the CP function.	X	-	-	X	-	Downlink Data Notification Delay
Suggested Buffering Packets Count	C	This IE may be present if the UP Function indicated support of the feature UDBC. When present, it shall contain the number of packets that are suggested to be buffered when the Apply Action parameter requests to buffer the packets. The packets that exceed the limit shall be discarded.		X	X	X	-	Suggested Buffering Packets Count
MT-EDT Control Information	O	This IE may be included to request the SGW-U to report the sum of DL Data Packets Size.	X	-	-	-	-	MT-EDT Control Information

7.5.4.12 Remove BAR IE within PFCP Session Modification Request

The Remove BAR grouped IE shall be encoded as shown in Figure 7.5.4.12-1.

Table 7.5.4.12-1: Remove BAR IE within PFCP Session Modification Request

Octet 1 and 2	Remove BAR IE Type = 87 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
BAR ID	M	This IE shall identify the BAR to be deleted.	X	-	-	X	-	BAR ID

7.5.4.13 Update Traffic Endpoint IE within PFCP Session Modification Request

The Update Traffic Endpoint grouped IE shall be encoded as shown in Figure 7.5.4.13-1.

Table 7.5.4.13-1: Update Traffic Endpoint IE within PFCP Session Modification Request

Octet 1 and 2	Update Traffic Endpoint Type = 129 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	

Traffic Endpoint ID	M	This IE shall uniquely identify the Traffic Endpoint to be modified for that PFCP session.	X	X	X	X	X	Traffic Endpoint ID
Local F-TEID	C	This IE shall be present if it needs to be changed. The CP function shall set the CHOOSE (CH) bit to 1 if the CP function requests the UP function to assign a local F-TEID to the PDR. See NOTE.	X	-	-	X	-	F-TEID
Network Instance	O	If present, this IE shall identify the Network instance to match for the incoming packet. See NOTE.	X	X	X	X	X	Network Instance
Redundant Transmission Detection Parameters	C	This IE shall be present if it needs to be changed See Table 7.5.2.2-5. See NOTE.	-	-	-	X	-	Redundant Transmission Detection Parameters
UE IP address	C	This IE shall be present if it needs to be changed. In the 5GC, several IEs with the same IE type may be present to represent multiple UE IP addresses, if the UPF indicated support of the IP6PL feature (see clause 5.21). When present, the UE IP address(es) present in this IE shall replace the UE IP address(es) stored in the UP function for this traffic endpoint. See NOTE.	-	X	X	X	-	UE IP address
Framed-Route	C	This IE shall be present for a PDR if the UPF indicated support of Framed Routing (see clauses 8.2.25 and 5.16) and it needs to be changed. If present, this IE shall describe a framed route. Several IEs with the same IE type may be present to provision a list of framed routes.	-	X	-	X	-	Framed-Route
Framed-Routing	C	This IE shall be present for a DL PDR if the UPF indicated support of Framed Routing (see clauses 8.2.25 and 5.16) and it needs to be changed. If present, this IE shall describe the routing method for the UP function for the IP route related to Framed-Routes or Framed-IPv6-Routes. (NOTE 2)	-	X	-	X	-	Framed-Routing
Framed-IPv6-Route	C	This IE shall be present for a PDR if the UPF indicated support of Framed Routing (see clauses 8.2.25 and 5.16) and it needs to be changed. If present, this IE shall describe a framed IPv6 route. Several IEs with the same IE type may be present to provision a list of framed IPv6 routes.	-	X	-	X	-	Framed-IPv6-Route
QFI	C	This IE shall be present if QFI(s) applicable for the traffic endpoints need to be changed and if the UPF has indicated it supports MTE feature as specified in clause 8.2.25. If present, this IE shall identify the QoS Flow Identifier to match for the incoming packet. Several IEs with the same IE type may be present to provision a list of QFIs. When present, the full set of applicable QFIs shall be provided.	-	-	-	X	-	QFI
Source Interface Type	C	This IE shall be present if the 3GPP interface type of the traffic endpoint needs to be changed.	X	X	-	X	X	3GPP Interface Type
RAT Type	O	This IE may be present if there is a RAT change for the UL PDR(s) if the PFCP session is established for a MA PDU session.	-	-	-	X	-	RAT Type
NOTE 1: The IEs which do not need to be modified shall not be included in the Update Traffic Endpoint IE. The UP function shall continue to behave according to the values previously received for IEs not present in the Update Traffic Endpoint IE. F-TEID may be changed if the SGW-C has received the "Change F-TEID support Indication" over the S11/S4 interface (for an IDLE state UE initiated TAU/RAU procedure to allow the SGW changing the GTP-U F-TEID).								
NOTE 2: In this release of specification, the UP function shall announce the IP route(s) for Framed-Route(s) or Framed-IPv6-Route(s) to the PDN regardless of the value of the Framed-Routing.								

7.5.4.14 Remove Traffic Endpoint IE within PFCP Session Modification Request

The Remove Traffic Endpoint grouped IE shall be encoded as shown in Figure 7.5.4.14-1.

Table 7.5.4.14-1: Remove Traffic Endpoint IE within PFCP Session Modification Request

Octet 1 and 2		Remove Traffic Endpoint IE Type = 130 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
Traffic Endpoint ID	M	This IE shall identify the Traffic Endpoint to be deleted.	X	X	X	X	X	Traffic Endpoint ID

7.5.4.15 Remove MAR IE within PFCP Session Modification Request

The Remove MAR grouped IE shall be encoded as shown in Figure 7.5.4.15-1.

Table 7.5.4.15-1: Remove MAR IE within PFCP Session Modification Request

Octet 1 and 2		Remove MAR IE Type = 168 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
MAR ID	M	This IE shall identify the MAR to be deleted.	-	-	-	X	-	MAR ID

7.5.4.16 Update MAR IE within PFCP Session Modification Request

The Update MAR grouped IE shall be encoded as shown in Figure 7.5.4.16-1.

Table 7.5.4.16-1: Update MAR IE within PFCP Session Modification Request

Octet 1 and 2		Update MAR IE Type = 169 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	

MAR ID	M	This IE shall identify the MAR to be updated.	-	-	-	X	-	MAR ID
Steering Functionality	C	This IE shall be present if it is changed.	-	-	-	X	-	Steering Functionality
Steering Mode	C	This IE shall be present if it is changed.	-	-	-	X	-	Steering Mode
Update 3GPP Access Forwarding Action Information	C	This IE shall be present if the 3GPP Access Forwarding Action Information was provisioned previously and if any of IEs is to be changed. This IE shall also be present to remove 3GPP Access Forwarding Action Information that was provisioned previously if the UE deregisters from the corresponding access. This shall be done by including this IE with a null length.	-	-	-	X	-	Update 3GPP Access Forwarding Action Information
Update Non-3GPP Access Forwarding Action Information	C	This IE shall be present if the Non-3GPP Access Forwarding Action Information was provisioned previously and if any of IEs is to be changed. This IE shall also be present to remove the Non-3GPP Access Forwarding Action Information that was provisioned previously if the UE deregisters from the corresponding access. This shall be done by including this IE with a null length.	-	-	-	X	-	Update Non-3GPP Access Forwarding Action Information
3GPP Access Forwarding Action Information	C	This IE shall be present to provision 3GPP access specific forwarding action information when this access is added, i.e. when the UE registers to 3GPP access. See Table 7.5.2.8-2.	-	-	-	X	-	3GPP Access Forwarding Action Information
Non-3GPP Access Forwarding Action Information	C	This IE shall be present to provision Non-3GPP access specific forwarding action information when this access is added, i.e. when the UE registers to non-3GPP access. See Table 7.5.2.8-3.	-	-	-	X	-	Non-3GPP Access Forwarding Action Information
Threshold values	C	This IE shall be present if it is changed.	-	-	-	X	-	Thresholds
Steering Mode Indicator	C	This IE shall be included if any of the following flag is set to "1" or if the change of flag(s) from "1" to "0" results in the IE becoming set to all zeros. Applicable flags are: - ALBI (Autonomous Load Balancing Indicator): this flag shall be set to "1" if the SMF allows the UPF to apply autonomous load-balance when the Steering Mode is Load-Balancing; - UEAI (UE Assistance Indicator): this flag shall be set to "1" if the SMF allows UE assistant load-balance when the Steering Mode is Load-Balancing.	-	-	-	X	-	Steering Mode Indicator

Table 7.5.4.16-2: Update 3GPP Access Forwarding Action Information IE in Update MAR IE

Octet 1 and 2		Update 3GPP Access Forwarding Action Information IE Type = 175 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
FAR ID	C	This IE shall be present if it is changed.	-	-	-	X	-	FAR ID
Weight	C	This IE shall be present if it is changed.	-	-	-	X	-	Weight
Priority	C	This IE shall be present if it is changed.	-	-	-	X	-	Priority
URR ID	C	This IE shall be present if a measurement action shall be applied or no longer applied to packets for this access. When present, this IE shall contain the list of all the URR IDs to be associated to this access.	-	-	-	X	-	URR ID
RAT Type	O	This IE may be present if it is changed.	-	-	-	X	-	RAT Type

Table 7.5.4.16-3: Update Non-3GPP Access Forwarding Action Information IE in Update MAR IE

Octet 1 and 2	Update Non-3GPP Access Forwarding Action Information IE Type = 176 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
Same IEs and requirements as defined in Table 7.5.4.16-2								

7.5.4.17 Create PDR/FAR/URR/QER/BAR/MAR IEs within PFCP Session Modification Request

PFCP Session Modification Request message may conditionally include Create PDR IE, Create FAR IE, Create URR IE, Create QER IE, Create BAR IE and Create MAR IE. The encoding of these IEs are specified within PFCP Session Establishment Request message (see clauses 7.5.2.2 – 7.5.2.8).

7.5.4.18 TSC Management Information IE within PFCP Session Modification Request

The TSC Management Information grouped IE shall be encoded as shown in Table 7.5.4.18-1.

Table 7.5.4.18-1: TSC Management Information IE within PFCP Session Modification Request

Octet 1 and 2	TSC Management Information IE Type = 199 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
Port Management Information Container	O	When present, this IE shall contain a Port Management Information container.	-	-	-	X	-	Port Management Information Container
User Plane Node Management Information Container	O	When present, this IE shall contain an User Plane Node Management Information container.	-	-	-	X	-	User Plane Node Management Information Container
NW-TT Port Number	C	When PMIC IE is present, this IE shall contain the related NW-TT Port Number.	-	-	-	X	-	NW-TT Port Number

7.5.4.19 Remove SRR IE within PFCP Session Modification Request

The Remove SRR grouped IE shall be encoded as shown in Table 7.5.4.19-1.

Table 7.5.4.19-1: Remove SRR IE within PFCP Session Modification Request

Octet 1 and 2	Remove SRR IE Type = 211 (decimal)							
Octets 3 and 4	Length = n							
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
SRR ID	M	This IE shall identify the SRR to be deleted.	-	-	-	X	-	SRR ID

7.5.4.20 Update SRR IE within PFCP Session Modification Request

The Update SRR grouped IE shall be encoded as shown in Table 7.5.4.20-1.

Table 7.5.4.20-1: Update SRR IE within PFCP Session Modification Request

Octet 1 and 2		Update SRR IE Type = 213 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
SRR ID	M	This IE shall uniquely identify the SRR among all the SRRs configured for that PFCP session	-	-	-	X	-	SRR ID
Access Availability Control Information	C	This IE shall be present if the Access Availability Control Information needs to be modified. See Table 7.5.2.9-2. The CP function shall provide the full Access Availability Control Information IE. The UP function shall replace the Access Availability Control Information IE provisioned earlier, if any, by the new received IE.	-	-	-	X	-	Access Availability Control Information
QoS Monitoring per QoS flow Control Information	C	This IE shall be present if the QoS Monitoring per QoS flow Control Information needs to be modified. See Table 7.5.2.9-3. The CP function shall provide the full set of QoS Monitoring per QoS flow Control Information IE(s). The UP function shall replace any QoS Monitoring per QoS flow Control Information IE(s) provisioned earlier by the new set of received IE(s). Several IEs within the same IE type may be present to represent a list of QoS Monitoring per QoS flow Control Information for different QoS flows.	-	-	-	X	-	QoS Monitoring per QoS flow Control Information
Direct Reporting Information	C	This IE shall be present if the Direct Reporting Information IE needs to be modified. See Table 7.5.2.9-4.	-	-	-	X	-	Direct Reporting Information

7.5.4.21 Ethernet Context Information within PFCP Session Modification Request

The Ethernet Context Information grouped IE shall be encoded as shown in Table 7.5.4.21-1.

Table 7.5.4.21-1: Ethernet Context Information IE within PFCP Session Modification Request

Octet 1 and 2		Ethernet Context Information IE Type = 254 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
MAC Addresses Detected	M	This IE shall be present if one or more MAC addresses need to be associated to the PDU session. Several IEs with the same IE type may be present to provision multiple lists of MAC addresses (e.g. with different V-LAN tags).	-	-	-	X	-	MAC Addressed Detected

7.5.4.22 Query Packet Rate Status IE within PFCP Session Modification Request

The Query Packet Rate Status grouped IE shall be encoded as shown in Table 7.5.4.22-1.

Table 7.5.4.22-1: Query Packet Rate Status IE within PFCP Session Modification Request

Octet 1 and 2		Query Packet Rate Status IE Type = 263 (decimal)						
Octets 3 and 4		Length = n						
Information elements	P	Condition / Comment	Appl.					IE Type
			Sx a	Sx b	Sx c	N4	N4 mb	
QER ID	M	This IE shall identify the QER being queried for its Packet Rate Status	-	X	-	X	-	QER ID

124	QFI	Extendable / Clause 8.2.89	1
125	Query URR Reference	Extendable / Clause 8.2.90	4
126	Additional Usage Reports Information	Extendable / Clause 8.2.91	2
127	Create Traffic Endpoint	Extendable / Table 7.5.2.7	Not Applicable
128	Created Traffic Endpoint	Extendable / Table 7.5.3.5	Not Applicable
129	Update Traffic Endpoint	Extendable / Table 7.5.4.13	Not Applicable
130	Remove Traffic Endpoint	Extendable / Table 7.5.4.14	Not Applicable
131	Traffic Endpoint ID	Extendable / Clause 8.2.92	1
132	Ethernet Packet Filter	Extendable / Table 7.5.2.2-3	Not Applicable
133	MAC address	Extendable / Clause 8.2.93	1
134	C-TAG	Extendable / Clause 8.2.94	3
135	S-TAG	Extendable / Clause 8.2.95	3
136	Ethertype	Extendable / Clause 8.2.96	2
137	Proxying	Extendable / Clause 8.2.97	1
138	Ethernet Filter ID	Extendable / Clause 8.2.98	4
139	Ethernet Filter Properties	Extendable / Clause 8.2.99	1
140	Suggested Buffering Packets Count	Extendable / Clause 8.2.100	1
141	User ID	Extendable / Clause 8.2.101	1
142	Ethernet PDU Session Information	Extendable / Clause 8.2.102	1
143	Ethernet Traffic Information	Extendable / Table 7.5.8.3-3	Not Applicable
144	MAC Addresses Detected	Extendable / Clause 8.2.103	7
145	MAC Addresses Removed	Extendable / Clause 8.2.104	7
146	Ethernet Inactivity Timer	Extendable / Clause 8.2.105	4
147	Additional Monitoring Time	Extendable / Table 7.5.2.4-3	Not Applicable
148	Event Quota	Extendable / Clause 8.2.112	4
149	Event Threshold	Extendable / Clause 8.2.113	4
150	Subsequent Event Quota	Extendable / Clause 8.2.106	4
151	Subsequent Event Threshold	Extendable / Clause 8.2.107	4
152	Trace Information	Extendable / Clause 8.2.108	7
153	Framed-Route	Variable Length / Clause 8.2.109	Not Applicable
154	Framed-Routing	Fixed Length / Clause 8.2.110	4
155	Framed-IPv6-Route	Variable Length / Clause 8.2.111	Not Applicable
156	Time Stamp	Extendable / Clause 8.2.114	4
157	Averaging Window	Extendable / Clause 8.2.115	4
158	Paging Policy Indicator	Extendable / Clause 8.2.116	1
159	APN/DNN	Variable Length / Clause 8.2.117	Not Applicable
160	3GPP Interface Type	Extendable / Clause 8.2.118	1
161	PFCPSRReq-Flags	Extendable / Clause 8.2.119	1
162	PFCPAUReq-Flags	Extendable / Clause 8.2.120	1
163	Activation Time	Extendable / Clause 8.2.121	4
164	Deactivation Time	Extendable / Clause 8.2.122	4
165	Create MAR	Extendable / Table 7.5.2.8-1	Not Applicable
166	3GPP Access Forwarding Action Information	Extendable / Table 7.5.2.8-2	Not Applicable
167	Non-3GPP Access Forwarding Action Information	Extendable / Table 7.5.2.8-3	Not Applicable
168	Remove MAR	Extendable / Table 7.5.4.15-1	Not Applicable
169	Update MAR	Extendable / Table 7.5.4.16-1	Not Applicable
170	MAR ID	Extendable / Clause 8.2.123	2
171	Steering Functionality	Extendable / Clause 8.2.124	1
172	Steering Mode	Extendable / Clause 8.2.125	1
173	Weight	Fixed / Clause 8.2.126	1
174	Priority	Extendable / Clause 8.2.127	1
175	Update 3GPP Access Forwarding Action Information	Extendable / Table 7.5.4.16-2	Not Applicable
176	Update Non 3GPP Access Forwarding Action Information	Extendable / Table 7.5.4.16-3	Not Applicable
177	UE IP address Pool Identity	Extendable / Clause 8.2.128	2
178	Alternative SMF IP Address	Extendable / Clause 8.2.129	1
179	Packet Replication and Detection Carry-On Information	Extendable / Clause 8.2.130	1
180	SMF Set ID	Extendable / Clause 8.2.131	Not applicable
181	Quota Validity Time	Extendable / Clause 8.2.132	4
182	Number of Reports	Fixed / Clause 8.2.133	2

183	PFCP Session Retention Information (within PFCP Association Setup Request)	Extendable / Table 7.4.4.1-2	1
184	PFCPASRsp-Flags	Extendable / Clause 8.2.134	1
185	CP PFCP Entity IP Address	Extendable / Clause 8.2.135	1
186	PFCPSReq-Flags	Extendable / Clause 8.2.136	1
187	User Plane Path Recovery Report	Extendable / Table 7.4.5.1.3-1	Not Applicable
188	IP Multicast Addressing Info within PFCP Session Establishment Request	Extendable / Clause 7.5.2.2-4	Not Applicable
189	Join IP Multicast Information IE within Usage Report	Extendable / Table 7.5.8.3-4	Not Applicable
190	Leave IP Multicast Information IE within Usage Report	Extendable / Table 7.5.8.3-5	Not Applicable
191	IP Multicast Address	Extendable / Clause 8.2.137	1
192	Source IP Address	Extendable / Clause 8.2.138	1
193	Packet Rate Status	Extendable / Clause 8.2.139	1
194	Create Bridge Info for TSC	Extendable / Clause 8.2.140	1
195	Created Bridge Info for TSC	Extendable / Table 7.5.3.6-1	Not Applicable
196	DS-TT Port Number	Fixed Length / Clause 8.2.141	4
197	NW-TT Port Number	Fixed Length / Clause 8.2.142	4
198	5GS User Plane Node	Extendable / Clause 8.2.143	Not Applicable
199	TSC Management Information IE within PFCP Session Modification Request	Extendable / Table 7.5.4.18-1	Not Applicable
200	TSC Management Information IE within PFCP Session Modification Response	Extendable / Table 7.5.5.3-1	Not Applicable
201	TSC Management Information IE within PFCP Session Report Request	Extendable / Table 7.5.8.5-1	Not Applicable
202	Port Management Information Container	Variable Length / Clause 8.2.144	Not Applicable
203	Clock Drift Control Information	Extendable / Table 7.4.4.1.2-1	Not Applicable
204	Requested Clock Drift Information	Extendable / Clause 8.2.145	1
205	Clock Drift Report	Extendable / Table 7.4.5.1.4-1	Not Applicable
206	Time Domain Number	Extendable / Clause 8.2.146	1
207	Time Offset Threshold	Extendable / Clause 8.2.147	8
208	Cumulative rateRatio Threshold	Extendable / Clause 8.2.148	4
209	Time Offset Measurement	Extendable / Clause 8.2.149	8
210	Cumulative rateRatio Measurement	Extendable / Clause 8.2.150	4
211	Remove SRR	Extendable/ Table 7.5.4.19-1	Not applicable
212	Create SRR	Extendable/ Table 7.5.2.9-1	Not applicable
213	Update SRR	Extendable/ Table 7.5.4.21-1	Not applicable
214	Session Report	Extendable / Table 7.5.8.7-1	Not Applicable
215	SRR ID	Extendable / Clause 8.2.151	1
216	Access Availability Control Information	Extendable / Table 7.5.2.9-2	Not applicable
217	Requested Access Availability Information	Extendable / Clause 8.2.152	1
218	Access Availability Report	Extendable / Table 7.5.8.6-2	Not applicable
219	Access Availability Information	Extendable / Clause 8.2.153	1
220	Provide ATSSS Control Information	Extendable / Table 7.5.2.10-1	Not Applicable
221	ATSSS Control Parameters	Extendable / Table 7.5.3.7-1	Not Applicable
222	MPTCP Control Information	Extendable / Clause 8.2.154	1
223	ATSSS-LL Control Information	Extendable / Clause 8.2.155	1
224	PMF Control Information	Extendable / Clause 8.2.156	1
225	MPTCP Parameters	Extendable / Table 7.5.3.7-2	Not Applicable
226	ATSSS-LL Parameters	Extendable / Table 7.5.3.7-3	Not Applicable
227	PMF Parameters	Extendable / Table 7.5.3.7-4	Not Applicable
228	MPTCP Address Information	Extendable / Clause 8.2.157	4
229	UE Link-Specific IP Address	Extendable / Clause 8.2.158	1
230	PMF Address Information	Extendable / Clause 8.2.159	1
231	ATSSS-LL Information	Extendable / Clause 8.2.160	1
232	Data Network Access Identifier	Variable Length / Clause 8.2.161	Not applicable
233	UE IP address Pool Information	Extendable / Table 7.4.4.1-3	Not Applicable
234	Average Packet Delay	Extendable / Clause 8.2.162	4
235	Minimum Packet Delay	Extendable / Clause 8.2.163	4
236	Maximum Packet Delay	Extendable / Clause 8.2.164	4
237	QoS Report Trigger	Extendable / Clause 8.2.165	1
238	GTP-U Path QoS Control Information	Extendable / Table 7.4.4.1.3-1	Not Applicable

239	GTP-U Path QoS Report (PFCP Node Report Request)	Extendable / Table 7.4.5.1.5-1	Not Applicable
240	QoS Information in GTP-U Path QoS Report	Extendable / Table 7.4.5.1.6-1	Not Applicable
241	GTP-U Path Interface Type	Extendable / Clause 8.2.166	1
242	QoS Monitoring per QoS flow Control Information	Extendable / Table 7.5.2.9-3	Not applicable
243	Requested QoS Monitoring	Extendable / Clause 8.2.167	1
244	Reporting Frequency	Extendable / Clause 8.2.168	1
245	Packet Delay Thresholds	Extendable / Clause 8.2.169	1
246	Minimum Wait Time	Extendable / Clause 8.2.170	4
247	QoS Monitoring Report	Extendable / Table 7.5.8.6-3	Not applicable
248	QoS Monitoring Measurement	Extendable / Clause 8.2.171	1
249	MT-EDT Control Information	Extendable / Clause 8.2.172	1
250	DL Data Packets Size	Extendable / Clause 8.2.173	2
251	QER Control Indications	Extendable / Clause 8.2.174	1
252	Packet Rate Status Report	Extendable / Table 7.5.7.1-2	Not applicable
253	NF Instance ID	Fixed / Clause 8.2.175	16
254	Ethernet Context Information	Extendable / Table 7.5.4.21-1	Not Applicable
255	Redundant Transmission Parameters	Extendable / Table 7.5.2.2-5, Table 7.5.2.3-4	Not Applicable
256	Updated PDR	Extendable / Table 7.5.5.5-1	Not Applicable
257	S-NSSAI	Fixed Length / Clause 8.2.176	4
258	IP version	Extendable / Clause 8.2.177	1
259	PFCPASReq-Flags	Extendable / Clause 8.2.178	1
260	Data Status	Extendable / Clause 8.2.179	1
261	Provide RDS configuration information	Extendable / Table 7.5.2.11-1	Not Applicable
262	RDS configuration information	Extendable / Clause 8.2.180	1
263	Query Packet Rate Status IE within PFCP Session Modification Request	Extendable / Table 7.5.4.22-1	Not Applicable
264	Packet Rate Status Report IE within PFCP Session Modification Response	Extendable / Table 7.5.5.4-1	Not Applicable
265	MPTCP Applicable Indication	Extendable / Clause 8.2.181	1
266	Bridge Management Information Container	Variable Length / Clause 8.2.182	Not Applicable
267	UE IP Address Usage Information	Extendable / Table 7.4.4.3.1-1	Not Applicable
268	Number of UE IP Addresses	Extendable / Clause 8.2.183	1
269	Validity Timer	Extendable / Clause 8.2.184	2
270	Redundant Transmission Forwarding Parameters	Extendable / Table 7.5.2.3-4	Not Applicable
271	Transport Delay Reporting	Extendable / Table 7.5.2.2-6	Not Applicable
272	Partial Failure Information	Extendable / Table 7.5.3.1-2	Not Applicable
273	Spare		
274	Offending IE Information	Variable Length / Clause 8.2.152	Not Applicable
275	RAT Type	Extendable / Table 8.2.186	1
276	L2TP Tunnel Information	Extendable / Table 7.5.2.1-2	Not Applicable
277	L2TP Session Information	Extendable / Table 7.5.2.1-3	Not Applicable
278	L2TP User Authentication IE	Extendable / Clause 8.2.187	1
279	Created L2TP Session	Extendable / Table 7.5.3.1-3	Not Applicable
280	LNS Address	Extendable / Clause 8.2.188	1
281	Tunnel Preference	Fixed Length / Clause 8.2.189	3
282	Calling Number	Variable Length / Clause 8.2.190	Not Applicable
283	Called Number	Variable Length / Clause 8.2.191	Not Applicable
284	L2TP Session Indications	Extendable / Clause 8.2.192	1
285	DNS Server Address	Extendable / Clause 8.2.193	1
286	NBNS Server Address	Extendable / Clause 8.2.194	1
287	Maximum Receive Unit	Fixed Length / Clause 8.2.195	2
288	Thresholds	Extendable / Clause 8.2.196	1
289	Steering Mode Indicator	Extendable / Clause 8.2.197	1
290	PFCP Session Change Info	Extendable / Table 7.4.7.1-2	Not Applicable
291	Group ID	Variable Length / Clause 8.2.198	Not Applicable
292	CP IP Address	Variable / Clause 8.2.199	Not Applicable
293	IP Address and Port number Replacement	Extendable / Clause 8.2.200	1
294	DNS Query Filter	Extendable / Clause 8.2.201	2
295	Direct Reporting Information	Extendable / Table 7.5.2.9-4	Not applicable

Octets	Bits							
	8	7	6	5	4	3	2	1
1 to 2	Type = 61 (decimal)							
3 to 4	Length = n							
5	ADNP	AURL	AFD	DNP	CP	DN	URL	FD
6	Spare							
m to (m+1)	Length of Flow Description							
(m+2) to p	Flow Description							
q to (q+1)	Length of URL							
(q+2) to r	URL							
s to (s+1)	Length of Domain Name							
(s+2) to t	Domain Name							
u to (u+1)	Length of Custom PFD Content							
(u+2) to v	Custom PFD Content							
w to (w+1)	Length of Domain Name Protocol							
(w+2) to x	Domain Name Protocol							
y to (y+1)	Length of Additional Flow Description							
(y+2) to z	Additional Flow Description							
a to (a+1)	Length of Additional URL							
(a+2) to b	Additional URL							
c to (c+1)	Length of Additional Domain Name and Domain Name Protocol							
(c+2) to d	Additional Domain Name and Domain Name Protocol							
e to (n+4)	These octet(s) is/are present only if explicitly specified							

Figure 8.2.39-1: PFD Contents

The following flags are coded within Octet 5 in the Figure 8.2.39-1:

- Bit 1 – FD (Flow Description): If this bit is set to "1", then the Length of Flow Description and the Flow Description fields shall be present, otherwise they shall not be present.
- Bit 2 – URL (URL): If this bit is set to "1", then the Length of URL and the URL fields shall be present, otherwise they shall not be present.
- Bit 3 – DN (Domain Name): If this bit is set to "1", then the Length of Domain Name and the Domain Name fields shall be present, otherwise they shall not be present.
- Bit 4 – CP (Custom PFD Content): If this bit is set to "1", then the Length of Custom PFD Content and the Custom PFD Content fields shall be present, otherwise they shall not be present.
- Bit 5 – DNP (Domain Name Protocol): If this bit is set to "1", then the Length of Domain Name Protocol and the Domain Name Protocol shall be present, otherwise they shall not be present; and if this bit is set to "1", the Length of Domain Name and the Domain Name fields shall also be present.
- Bit 6 – AFD (Additional Flow Description): If this bit is set to "1", the Length of Additional Flow Description and the Additional Flow Description field shall be present, otherwise they shall not be present.
- Bit 7 – AURL (Additional URL): If this bit is set to "1", the Length of Additional URL and the Additional URL field shall be present, otherwise they shall not be present.
- Bit 8 – ADNP (Additional Domain Name and Domain Name Protocol): If this bit is set to "1", the Length of Additional Domain Name and Domain Name Protocol, and the Additional Domain Name and Domain Name Protocol field shall be present, otherwise they shall not be present.

The Flow Description field, when present, shall be encoded as an OctetString as specified in clause 6.4.3.7 of 3GPP TS 29.251 [21].

The Domain Name field, when present, shall be encoded as an OctetString as specified in clause 6.4.3.9 of 3GPP TS 29.251 [21].

The URL field, when present, shall be encoded as an OctetString as specified in clause 6.4.3.8 of 3GPP TS 29.251 [21].

The Domain Name Protocol field, when present, shall be encoded as an OctetString as specified in clause 6.4.3.9 of 3GPP TS 29.251 [21].

Octets 5 to 10 represent the Trace Reference parameter as defined in clause 5.6 of 3GPP TS 32.422 [35].

Triggering Events, Session Trace Depth, List of Interfaces and IP Address of Trace Collection Entity are specified in 3GPP TS 32.422 [35].

Octets "(p+2) to q" shall contain an IPv4 address value (4 octets) or IPv6 address value (16 octets).

See 3GPP TS 24.008 [5], clause 10.5.1.4, Mobile Identity, for the coding of MCC and MNC. If MNC is 2 digits long, bits 5 to 8 of octet 6 are coded as "1111".

8.2.109 Framed-Route

The Framed-Route IE describes a framed route. It shall be encoded as shown in Figure 8.2.109-1.

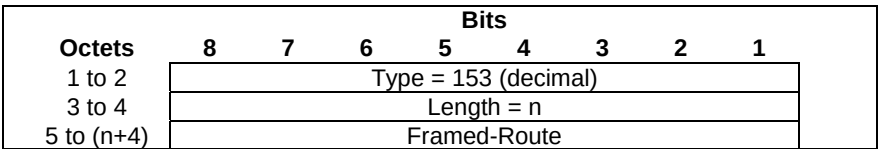


Figure 8.2.109-1: Framed-Route

The Framed-Route field shall be encoded as an Octet String as the value part of the Framed-Routing AVP specified in IETF RFC 2865 [37].

8.2.110 Framed-Routing

The Framed-Routing IE describes the frame routing of a framed route. It shall be encoded as shown in Figure 8.2.110-1.

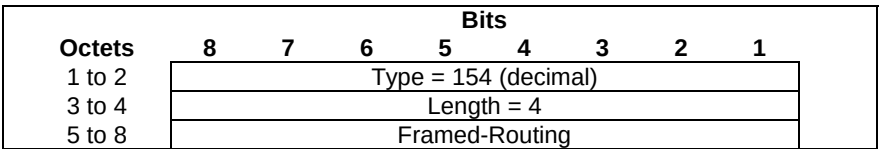


Figure 8.2.110-1: Framed-Routing

The Framed-Routing field shall be encoded as the value part of the Framed-Routing AVP specified in IETF RFC 2865 [37].

8.2.111 Framed-IPv6-Route

The Framed-IPv6-Route IE describes a framed IPv6 route. It shall be encoded as shown in Figure 8.2.111-1.

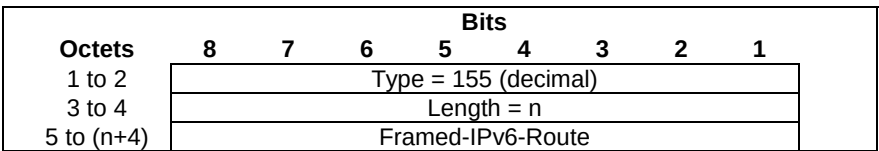


Figure 8.2.111-1: Framed-IPv6-Route

The Framed-IPv6-Route field shall be encoded as an Octet String as the value part of the Framed-IPv6-Route AVP specified in IETF RFC 3162 [38].

8.2.112 Event Quota

The Event Quota IE type shall be encoded as shown in Figure 8.2.112-1. It contains the event quota to be monitored by the UP function.

Octets	Bits							
	8	7	6	5	4	3	2	1
1 to 2	Type = 191 (decimal)							
3 to 4	Length = n							
5	Spare				A	R	V4	V6
m to (m+3)	(start) IPv4 address							
p to (p+15)	(start) IPv6 address							
q to (q+3)	(end) IPv4 address							
r to (r+15)	(end) IPv6 address							
k to (n+4)	These octet(s) is/are present only if explicitly specified							

Figure 8.2.137-1: IP Multicast Address

The following flags are coded within Octet 5:

- Bit 1 – V6: If this bit is set to "1", then the (start) IPv6 address field shall be present, otherwise the (start) IPv6 address field shall not be present.
- Bit 2 – V4: If this bit is set to "1", then the (start) IPv4 address field shall be present, otherwise the (start) IPv4 address field shall not be present.
- Bit 3 – R (Range): If this bit is set to "1", this indicates that a range of addresses is included, i.e. that
 - the (start) IPv4 address and (end) IPv4 address fields shall be present if bit 2 (V4) is set to "1";
 - the (start) IPv6 address and (end) IPv6 address fields shall be present if bit 1 (V6) is set to "1",

otherwise (end) address fields shall not be present.

- Bit 4 – Any: If this bit is set to "1", this indicates any IP multicast address; in this case, no IP address field shall be included.
- Bit 5 to 8 Spare, for future use and set to "0".

One and only one of the V6, V4 and A flags shall be set to "1". The R flag may be set if the V6 or the V4 flag is set to "1".

Octets "m to (m+3)", "p to (p+15)", "q to (q+3)", "r to (r+15)" (IPv4 address / IPv6 address fields), if present, shall contain the address value.

8.2.138 Source IP Address

The Source IP Address IE type shall be encoded as shown in Figure 8.2.138-1

Octets	Bits							
	8	7	6	5	4	3	2	1
1 to 2	Type = 192 (decimal)							
3 to 4	Length = n							
5	Spare				MPL	V4	V6	
m to (m+3)	IPv4 address							
p to (p+15)	IPv6 address							
q	mask/prefix length							
k to (n+4)	These octet(s) is/are present only if explicitly specified							

Figure 8.2.138-1: Source IP Address

The following flags are coded within Octet 5:

- Bit 1 – V6: If this bit is set to "1", then the IPv6 address field shall be present, otherwise the IPv6 address field shall not be present.

Octets	Bits							
	8	7	6	5	4	3	2	1
1 to 2	Type = 229 (decimal)							
3 to 4	Length = n							
5	Spare				NV6	NV4	V6	V4
p to (p+3)	UE Link-Specific IPv4 Address for 3GPP Access							
q to (q+15)	UE Link-Specific IPv6 Address for 3GPP Access							
r to (r+3)	UE Link-Specific IPv4 Address for Non-3GPP Access							
s to (s+3)	UE Link-Specific IPv6 Address for Non-3GPP Access							
m to (n+4)	These octet(s) is/are present only if explicitly specified							

Figure 8.2.158-1: UE Link-Specific IP Address

The following flags are coded within Octet 5:

- Bit 1 – V4: If this bit is set to "1", then the UE Link-Specific IPv4 Address for 3GPP Access shall be present in the UE Link-Specific IP Address IE.
- Bit 2 – V6: If this bit is set to "1", then the UE Link-Specific IPv6 Address for 3GPP Access shall be present in the UE Link-Specific IP Address IE.
- Bit 3 – NV4: If this bit is set to "1", then the UE Link-Specific IPv4 Address for Non-3GPP Access shall be present in the UE Link-Specific IP Address IE.
- Bit 4 – NV6: If this bit is set to "1", then the UE Link-Specific IPv6 Address for Non-3GPP Access shall be present in the UE Link-Specific IP Address IE.
- Bit 5 to 8 Spare, for future use and set to "0".

Octets "p to (p+3)" or "q to (q+15)" (IPv4 address / IPv6 address fields), if present, shall contain the value for UE Link-Specific IP Address for 3GPP access.

Octets "r to (r+3)" or "s to (s+15)" (IPv4 address / IPv6 address fields), if present, shall contain the value for UE Link-Specific IP Address for Non-3GPP access.

8.2.159 PMF Address Information

The PMF Address Information IE shall contain the address information of Performance Measure Function (PMF). It shall be encoded as shown in Figure 8.2.159-1.

	Bits							
Octets	8	7	6	5	4	3	2	1
1 to 2	Type = 230 (decimal)							
3 to 4	Length = n							
5	Spare					MAC	V6	V4
p to (p+3)	PMF IPv4 Address							
q to (q+15)	PMF IPv6 Address							
r to (r+1)	PMF Port for 3GPP Access							
s to (s+1)	PMF Port for Non-3GPP Access							
t to (t+5)	PMF MAC Address for 3GPP Access							
u to (u+5)	PMF MAC Address for Non-3GPP Access							
m to (n+4)	These octet(s) is/are present only if explicitly specified							

Figure 8.2.159-1: PMF Address Information

The following flags are coded within Octet 5:

- Bit 1 – V4: If this bit is set to "1", it indicates the PMF IPv4 Address field shall be present in the PMF Address Information IE, together with: PMF Port for 3GPP Access, PMF Port for Non-3GPP Access.
- Bit 2 – V6: If this bit is set to "1", it indicates the PMF IPv6 Address field shall be present in the PMF Address Information IE, together with: PMF Port for 3GPP Access, PMF Port for Non-3GPP Access.